

Customer Shopping Behavior Analysis

1. Project Overview:

The project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

- Rows : 3,900
- Columns : 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data : 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading :** Imported the datasets using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Customer ID           3900 non-null   int64
 1   Age                   3900 non-null   int64
 2   Gender                3900 non-null   object
 3   Item Purchased        3900 non-null   object
 4   Category              3900 non-null   object
 5   Purchase Amount (USD) 3900 non-null   int64
 6   Location              3900 non-null   object
 7   Size                  3900 non-null   object
 8   Color                 3900 non-null   object
 9   Season                3900 non-null   object
10   Review Rating         3863 non-null   float64
11   Subscription Status    3900 non-null   object
12   Shipping Type          3900 non-null   object
13   Discount Applied       3900 non-null   object
14   Promo Code Used        3900 non-null   object
15   Previous Purchases     3900 non-null   int64
16   Payment Method         3900 non-null   object
17   Frequency of Purchases 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization :** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering :**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.

Data Consistency Check : Verified if **discount_applied** and **promo_code_usd** were redundant ; dropped **promo_code_usd**.

Database Integration : Connected Python script to mysql-connector-python and loaded the cleaned DataFrame into the database for SQL analysis.

4.Data Analysis using SQL

We performed structured analysis in mysql-connector-python to answer key buisness questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Female	75191
	Male	157890

2. **High – Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount

Result Grid			
	discount_applied	customer_id	purchase_amount_usd
▶	Yes	1	53
	Yes	2	64
	Yes	3	73
	Yes	4	90
	Yes	5	49

3. **Highest average rating** – Identified 5 products which have highest rating review.

	item_purchased	avg_rating
▶	Gloves	3.8614285714285725
	Sandals	3.8443750000000003
	Boots	3.8187500000000005
	Hat	3.8012987012987005
	Skirt	3.784810126582278

4. **Comparison Between Subscribers and Non-Subscribers** – Compared Total revenue and average amount between Subscribers and non-subscribers.

Result Grid				
	total_customer	avg_amount	subscription_status	total_revenue
▶	554931	59.4919	Yes	62645
	7052019	59.8651	No	170436

5. **Comparison Between Standard and Express Shipping** – Identified the average Purchase Amounts between Standard and Express Shipping.

Result Grid		
	shipping_type	avg_amount
▶	Express	60.4752
	Standard	58.4602

6. **Top Products With Discount** - Identified 5 products have the highest percentage of purchases with discounts applied.

Result Grid   Filter Rows:		
	item_purchased	discount_percentage
▶	Hat	50.0000
	Sneakers	49.6552
	Coat	49.0683
	Sweater	48.1707
	Pants	47.3684

7. **Creating segments of customer** – Splitted customers in New , Loyal and Returning based on their previous purchases.

Result Grid   Filter Rows:		
	customer_segment	total_customers
▶	Loyal	3116
	Returning	701
	New	83

8. **Top Products** – Identified top 3 purchased product by each category.

Result Grid   Filter Rows:			
	category	item_purchased	total_purchases
▶	Accessories	Jewelry	171
	Accessories	Sunglasses	161
	Accessories	Belt	161
	Clothing	Blouse	171
	Clothing	Pants	171
	Clothing	Shirt	169
	Footwear	Sandals	160
	Footwear	Shoes	150
	Footwear	Sneakers	145
	Outerwear	Jacket	163
	Outerwear	Coat	161

9. **Customer who would want to subscribe**- Identified the customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe.

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	customer_type	total_customers	subscribed_customers	subscription_rate_percentage
▶	Repeat Buyer	3476	958	27.56
	Non-Repeat Buyer	424	95	22.41

10. Revenue Contributions – Identified revenue contribution of each group.

Result Grid			Filter Rows:
	total_amount	age_group	
▶	55763	Senior	
	55978	Adult	
	59197	Middle-aged	
	62143	Young Adult	



5. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

- **Customer Loyalty Programs** – Reward repeat buyers to move them into the ‘Loyal’ segment .
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best selling products in campaigns.
- **Trageted Marketing** – Focus efforts on high-revenue age groups and express-shipping.